Responsiveness of the Revised Low Vision Independence Measure (LVIM–R)

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Importance: The Low Vision Independence Measure (LVIM) was designed to measure the effectiveness of occupational therapy to improve visual ability in low vision rehabilitation.

Objective: To validate the Revised LVIM (LVIM–R) as an outcome measure by determining its sensitivity to visual ability changes after occupational therapy.

Design: In this observational study, LVIM–R scores were collected before and after customary low vision intervention.

Setting: Home health agency or outpatient facility.

Participants: Forty-four participants with a mean age of 80.2 yr (standard deviation = 11.2) and an average length of visual impairment of 6.2 yr. Twenty-three participants (52.3%) were treated in an outpatient setting, and 21 (47.7%) were treated in the home. The majority of participants were non-Hispanic White (92.8%) and had macular degeneration (54.5%).

Intervention: Customary low vision intervention by occupational therapists.

Outcomes and Measures: LVIM–R scores were collected before and after intervention, and participants' pre- and posttest scores were anchored into Rasch-calibrated item parameters.

Results: Paired *t* tests demonstrated significant increases in person measures with large effect sizes for both constructs of the LVIM–R, the visual field or scotoma ($t[43] = 6.46$, $p < .0001$; Cohen's $d = 0.92$) and visual acuity ($t[43] = 9.08$, $p < .0001$; Cohen's $d = 1.14$) constructs.

Conclusions and Relevance: The LVIM–R is sensitive to changes in visual ability in clients who have good rehabilitation potential and can be used to examine the effectiveness of occupational therapy for low vision.

What This Article Adds: The LVIM–R is useful for measuring occupational therapy outcomes in low vision rehabilitation in the home or in outpatient settings.

Approximately 3.22 million people in the United States had low vision in 2015, and this number is expected to increase to 6.95 million by 2050 (Varma et al., 2016). Low vision can significantly limit a person's ability to perform activities of daily living (ADLs) and instrumental activities of daily living (IADLs; Crews & Campbell, 2001) even with best correction (American Optometric Association, 2019). Activities commonly affected include reading, financial management, meal preparation, medication management, home management, leisure participation, functional mobility, out-of-home mobility, and driving (Brown et al., 2014; Crews & Campbell, 2001; Liu et al., 2013).

Visual impairments reduce visual ability, and the goal of low vision rehabilitation is to increase clients' ability to perform vision-dependent ADLs and IADLs. Low vision rehabilitation services have been shown to improve clients' clinical and functional ability outcomes (Binns et al., 2012). The extent of improvement has ranged from small to significant (Binns et al., 2012; Liu et al., 2013), and the variance in treatment effectiveness has been attributed in part to differences in intervention approaches and outcome measures used (Liu et al., 2013).

To assess the effectiveness of low vision rehabilitation, outcome measures must be accurate and precise and, to be clinically meaningful, they must be responsive to change. Instruments to measure outcomes of low vision rehabilitation must target the occupations and goals normally included in the treatment plans of clients with

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low vision. The Low Vision Independence Measure (LVIM) was designed to measure the visual ability of clients with a variety of diagnoses resulting in low vision (Smith, 2013). *Visual ability* is the "overall ability to perform activities that depend on vision" (Goldstein et al., 2014, p. 1170). When the LVIM was initially developed, many existing outcome measures for vision-dependent activities had been developed for a specific diagnosis, such as macular degeneration (Dahlin-Ivanoff et al., 2001), or focused heavily on performance skills such as reading (Haymes et al., 2001). The National Eye Institute (NEI) Visual Function Questionnaire–51 (Mangione et al., 1998) and the NEI Visual Function Questionnaire–25 (Mangione et al., 2001), commonly used at that time, address one vision-dependent activity, driving.

The LVIM was designed as a patient-reported outcome measure to support client-centered practice (Kyte et al., 2015) and aid in eliminating therapist bias from the assessment process (Wolfson et al., 2000). Although the number of client-reported instruments designed to measure visual ability outcomes has grown in recent years, many apply to medical treatment, including surgery for glaucoma, dry eye, refractive errors, amblyopia, cataract, macular diseases, and strabismus (Khadka et al., 2013). There remains a need for instruments that can be used to measure clients' responsiveness to low vision rehabilitation that addresses performance of a range of vision-dependent ADLs and IADLs. Assessments that contain items that may not need to be addressed in low vision rehabilitation, such as depression, could dilute measurement of the overall outcomes of the intervention. If constructs are measured for which the instrument was not designed, a null result can occur.

In a systematic review, Liu et al. (2013) found few studies that could demonstrate the effectiveness of occupational therapy in low vision rehabilitation, in part because some of the outcome measures conflated well-being with ADL and IADL performance. The American Occupational Therapy Association (AOTA; 2014a) listed the following intervention strategies for clients with low vision to maximize their independence and safety in preferred occupations: contrast enhancement, organizational strategies, lighting, glare reduction, filters to manage glare, sensory substitution, and magnification.

The LVIM was developed to measure outcomes of low vision rehabilitation targeted to improve occupational performance of vision-dependent ADLs and IADLs and increase participation (Smith, 2013). After initial development, expert feedback and the literature were used to confirm inclusion of problematic vision-dependent ADLs and IADLs. Items of the LVIM were organized into subscales that align with occupations named in the *Occupational Therapy Practice Framework: Domain and Process* (3rd ed.; *OTPF–3;* AOTA, 2014b) rather than being sorted into performance skills such as reading, visual–motor skills, and visual information, as is common in other visual function questionnaires (Massof et al., 2007). The focus of the LVIM on occupations is consistent with the domain of occupation therapy as outlined in the *OTPF–3:* the interaction of occupations, client factors, performance skills, performance patterns, and context and environment.

The LVIM was recently revised (LVIM–R), and its validity and reliability were established with the Rasch model as a measure of visual ability for both home health and outpatient clients with various causes of visual impairment (Smith et al., 2019). The instrument consists of two measurement constructs: (1) visual field or scotoma and (2) visual acuity. The purpose of the study described in this article was to validate the LVIM–R as an outcome measure by determining its responsiveness to changes after usual and customary occupational therapy interventions had been provided to clients with low vision.

Method

Design

In this observational study, collaborating occupational therapists administered the LVIM–R to clients before and after occupational therapy for low vision. We exercised no control over the specific occupational therapy interventions participants received.

Participants

Eight occupational therapists from six states, known by or referred to the first author (Theresa M. Smith), recruited participants from their caseload. The therapists' experience in low vision rehabilitation ranged from 1 to 12 yr (mean [*M*] = 5.2, standard deviation [*SD*] = 3.0). One of the therapists had earned special certification in low vision and certified low vision therapist credentials. Three had completed the Graduate Certificate in Low Vision Rehabilitation from the University of Alabama, Birmingham, and another had less than a year left to finish.

Clients who met the following inclusion criteria were asked by their treating therapist if they would be interested in participating in the study: (1) had been diagnosed with low vision because of an eye disease such as, but not limited to, diabetic retinopathy, age-related macular degeneration, cataract, or glaucoma; (2) was identified by the treating therapist as having good rehabilitation potential to meet plan-of-care goals established in collaboration with the client; and (3) was a new client of the treating therapist. Clients were excluded if they were currently receiving rehabilitation for any other health condition or were not fluent in English. The institutional review board of the authors' affiliated university approved the study and the informed consent form used to obtain consent before data collection.

Instruments

Revised Low Vision Independence Measure

The LVIM–R is a client-reported rating scale questionnaire designed to measure visual ability using difficulty ratings for vision-dependent ADLs and IADLs (Smith et al., 2019). It contains 46 items (activities) in eight subscales: Self-Care, Food Preparation, Home Management, Communication, Financial Management, Leisure, Shopping, and Mobility (Table 1). Level of difficulty for each activity is rated on a 4-point ordinal scale (1 = *unable,* 2 = *very difficult,* 3 = *moderately difficult,* and 4 = *not difficult*). If an activity is not relevant to a client, he or she can select *not applicable,* and the item response is not scored or is treated as missing data. Administration of the LVIM–R takes approximately 15 min.

The LVIM–R consists of two measurement constructs: (1) visual field or scotoma (24 items) and (2) visual acuity (22 items; Smith et al., 2019). Two of the Leisure items, "Reading books/newspapers/magazines" and "Using technology/iPads/e-books," can also be used for the Communication subscale. Fit statistics, item difficulty hierarchy, and differential item functioning for the visual field or scotoma and visual acuity constructs were detailed in a previous article (Smith et al., 2019).

Participant Trait Survey

Treating occupational therapists obtained data on participants' demographic characteristics, health history, and other traits using a standard interview survey. The survey included closed-ended questions about gender, age, diagnosis, comorbidities, and duration of the low vision condition.

Procedures

Data Collection

Collaborating occupational therapists were oriented to the study by the first author and provided with data forms with code number identifiers to maintain participant anonymity and conform to privacy requirements (Health Insurance Portability and Accountability Act of 1996, Pub. L. 104-191). In the first treatment session with each participant, the therapist read the questions aloud to complete the trait survey and then administered the LVIM–R pretest. The therapist then provided the participant with the course of treatment usually provided for clients with low vision. On average, clients were seen 1×/wk for 1 hour. LVIM–R posttest data were collected at discharge from occupational therapy services in either the participant's home or the outpatient clinic.

Table 1. Subscales and Items of the Revised Low Vision Independence Measure

Subscale	Items
Self-Care	Brushing teeth/cleaning dentures
	Identifying foods when eating
	Cutting food on plate
	Keeping food on utensil
	Managing medication
	Identifying clothing
Food Preparation	Reading recipes
	Pouring liquids
	Measuring food
	Slicing and peeling food
	Seasoning food
	Determining food doneness
	Using microwave
	Using stove
	Identifying cans
Home Management	Cleaning floors
	Cleaning counters
	Washing dishes
	Organizing cabinets
	Operating washer and dryer
Communication	Writing notes/letters/envelopes
	Signing your name
	Locating/dialing a phone number
	Documenting appointments
	Using computer/cell phone
Financial Management	Identifying coins and paper money
	Reading bills and receipts
	Writing checks
	Balancing accounts
	Banking online
Leisure	Watching television
	Playing games/cards
	Reading books/newspaper/magazines
	Using technology/iPads/e-books
	Doing crafts
	Eating out
	Participating in community
Shopping	Preparing a shopping list
	Locating items
	Reading labels or prices
	Determining expiration dates
	Paying for items
Mobility	Crossing streets
	Getting to appointments
	Visiting friends and family
	Using public transportation

Data Analysis

We generated descriptive statistics for participant traits using IBM SPSS Statistics Version 19 (IBM Corp., Armonk, NY). We performed a Rasch analysis with the Andrich (1979) rating scale model separately for the pre- and postintervention data using Winsteps Version 3.93.2 (Beaverton, OR).

Person Measure Estimations

Person measures for the pre- and posttests were obtained using information from the mean item difficulty and rating scale response categories with three-step thresholds generated on the basis of the LVIM–R validation study (Smith et al., 2019) to create the same measurement across the two time points. For example, the person measures of the visual field or scotoma and visual acuity constructs were obtained using the pre- and posttest raw scores based on the item parameters of the 24 items and 22 items, respectively (Wright, 1993). To conduct visual comparisons between the person measure estimations at pre- and posttest, we created item maps for the two constructs with the person measures and item difficulty measures on the same linear continuum (logit) across the two time points.

Changes in Visual Ability

Once the person measures at pre- and posttest for the two constructs were obtained, we ran separate paired *t* tests to examine the person measure changes between the time points. Effect size (ES; Cohen's *d*) for the magnitude of person measure changes was calculated by dividing the mean difference between pre- and posttest scores by the pooled standard deviation (Cohen, 1988).

ESs of $0.0 < d < 0.2$ were considered small, $0.3 < d < 0.5$ medium, and $d > 0.6$ large (Middel & van Sonderen, 2002). In addition, we calculated standardized response means (SRMs) as an ES index. SRM is the mean change in score divided by the standard deviation of the change score (Husted et al., 2000). SRMs of $0.0 < d < 0.2$ were considered small, $0.3 < d < 0.7$ medium, and $d > 0.8$ large (Middel & van Sonderen, 2002). Our sample size was small; many occupational therapists providing low vision rehabilitation do not see clients with low vision exclusively because the census of low vision clients can be much lower relative to clients with other diagnoses. Because of the small sample size, bootstrapping was used for 1,000 samples with replacement when estimating 95% confidence intervals (CIs) for the ESs and SRMs (Efron & Tibshirani, 1993).

Results

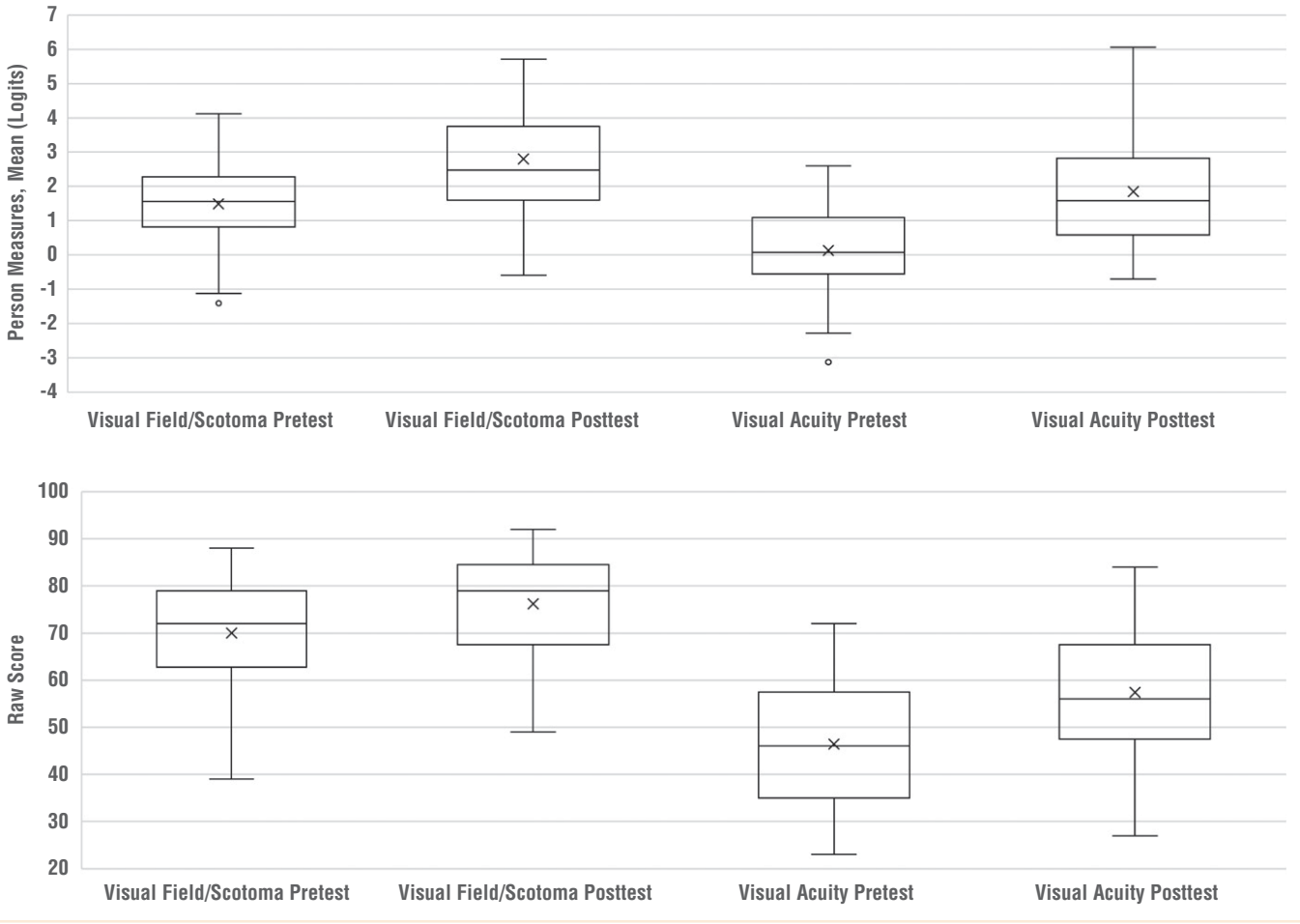
Participant Traits

A total of 44 participants completed the LVIM–R at both pre- and posttest. Twenty-three participants (52.3%) were treated in outpatient settings, and 21 (47.7%) received services from a home health agency. Participants included 11 men (25.0%), 32 women (72.7%), and 1 participant (2.3%) whose gender was not reported; ages ranged from 42 to 96 (M = 80.2, SD = 11.2). The majority of participants were non-Hispanic White (92.8%); 4.7% were Hispanic, and 2.4% were non-Hispanic Black. The average duration of visual impairment was 6.2 yr (SD = 5.4, range = 0.25–23.00). The most frequently reported primary visual disorder diagnosis was macular degeneration (n = 24), followed by cataract (n = 9), other (n = 5), glaucoma (n = 3), visual field deficit (n = 2), and diabetic retinopathy (n = 1).

Changes in Visual Ability

The mean person measure for the visual field or scotoma construct was 1.50 logits (SD = 1.11) at pretest and 2.80 logits (SD = 1.64) at posttest. The mean person measure for the visual acuity construct was 0.20 logits (SD = 1.20) at pretest and 1.72 logits (SD = 1.51) at posttest (Figure 1). Paired t tests demonstrated significantly increased person measures at posttest for visual field or scotoma, $\Delta = 1.30$, $t(43) = 6.46$, $p < .0001$, and visual acuity,

Figure 1. Pre- and posttest results for the visual field or scotoma and visual acuity constructs.



Δ = 1.52, $t(43) = 9.08$, $p < .0001$. These increases indicate large ESs for visual field or scotoma (d = 0.92, 95% CI [0.66, 1.18]; SRM = 0.97, 95% CI [0.58, 1.21]) and visual acuity (d = 1.14, 95% CI [0.93, 1.37]; SRM = 1.37, 95% CI [1.09, 1.63]).

We ran a sensitivity analysis stratifying the outpatient and home health settings to examine whether responsiveness to change was different by setting. No significant differences were found in pretest scores by setting for either visual field or scotoma, $t(42) = 0.86$, $p = .39$, or visual acuity, $t(42) = 1.04$, $p = .30$. Overall, the participants who received home health services demonstrated greater improvements (visual field or scotoma: d = 1.39, 95% CI [1.08, 1.86]; SRM = 1.57, 95% CI [1.16, 2.11]; visual acuity: d = 1.43, 95% CI [1.11, 1.81]; SRM = 1.91, 95% CI [1.44, 2.56]) compared with those who received outpatient services (visual field or scotoma: d = 0.52, 95% CI [0.16, 0.84]; SRM = 0.62, 95% CI [0.08, 0.84]; visual acuity: d = 0.87, 95% CI [0.62, 1.13]; SRM = 1.07, 95% CI [0.75, 1.36]).

Figures 2 and 3 present the person measure distributions between the pre- and posttests for the visual field or scotoma and visual acuity constructs with the person measures at pre- and posttest calibrated into the same linear interval scale (logits). Whereas 4 participants had person measure values greater than 3.0 at pretest on the visual field or scotoma construct (left panel in Figure 2), at posttest 18 participants' measures were greater than 3.0, and 9 had the maximum score (right panel in Figure 2). Similarly, whereas no participants had person measure values greater than 3.0 at pretest for the visual acuity construct (left panel in Figure 3), at posttest 9 participants had person measures greater than 3.0, and 3 had the maximum score (right panel in Figure 3).

Discussion

The aim of this study was to determine the responsiveness of the LVIM–R to change in visual ability after low vision rehabilitation. After participants received occupational therapy, the LVIM–R captured person measure differences for the constructs of visual field or scotoma and visual acuity; therefore, the LVIM–R can be considered responsive to change as a result of occupational therapy intervention for low vision in clients with good rehabilitation potential.

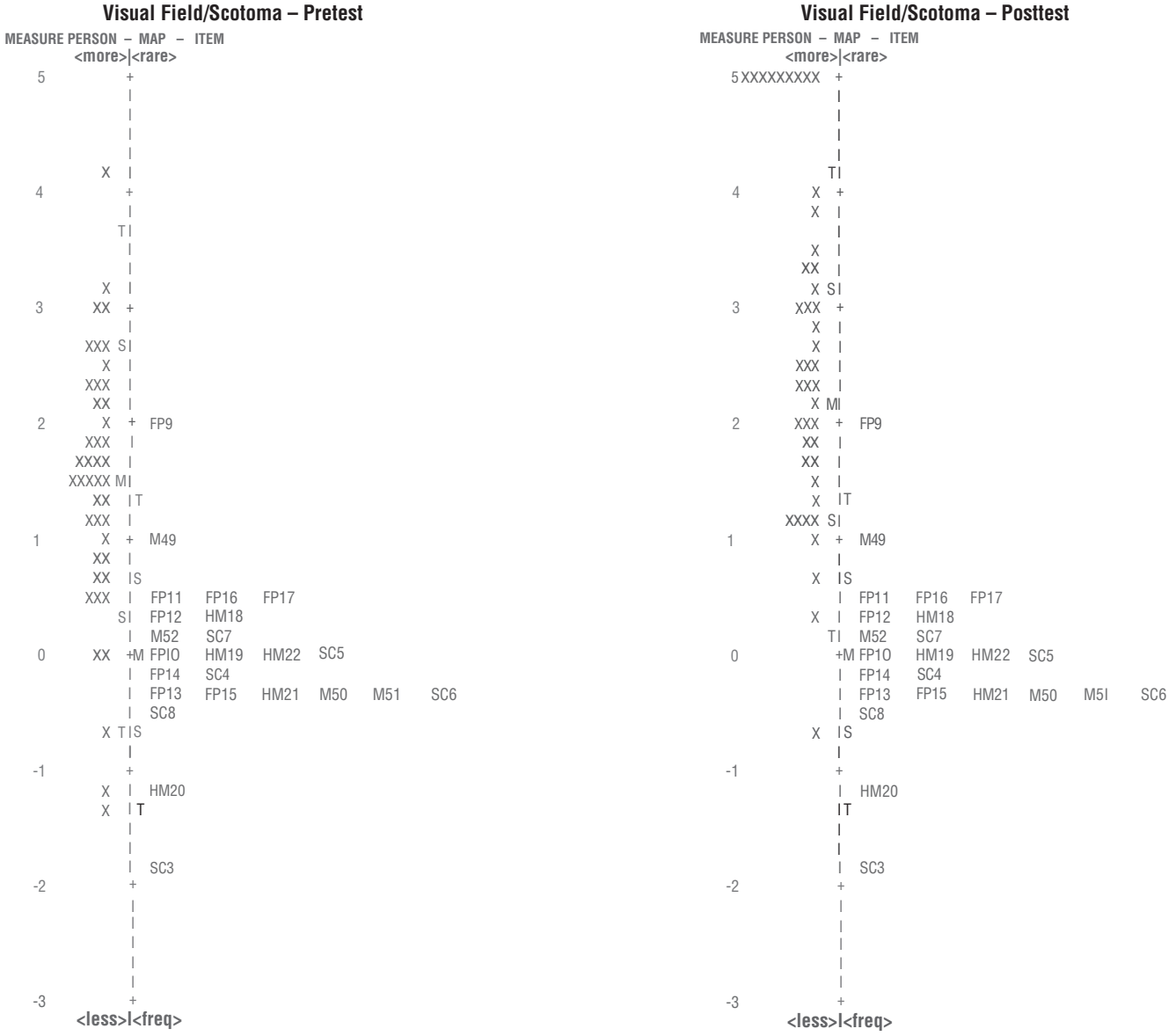
Stelmack et al. (2017) found that low vision rehabilitation was effective only for people with best corrected visual acuity worse than 20/63. Massof and Stelmack (2013) demonstrated that lower visual ability at baseline was associated with larger change scores and higher visual ability at baseline with smaller change scores; person measure differences varied by setting. We did not collect data on visual acuity but determined that participants who received low vision rehabilitation services in the home had greater person measure differences than those who received services in an outpatient setting. There may be clinical differences between homes and outpatient clinics, but we did not find any literature on instrument responsiveness related to these specific treatment settings.

At baseline, participants had difficulty with an average of 10 IADLs in the visual acuity construct. After occupational therapy, the mean person measure increased for these IADLs, meaning that participants had improved visual ability and experienced fewer difficulties with these tasks. Given the nature of the disease process in diagnoses that result in low vision, the expectation would be a decline in visual ability and not an increase. Improvement in this construct indicates that participants gained independence in IADLs after receiving occupational therapy.

Although participants also showed significant improvement in the visual field or scotoma construct, they had less difficulty with most of these items before receiving occupational therapy than with items in the visual acuity construct. This finding may be attributable in part to the large percentage of participants who had macular degeneration, which affects the central vision needed for visual acuity (Berdeaux et al., 2005). The two constructs need to be measured together to accurately capture visual ability improvements from occupational therapy.

Other researchers have addressed the responsiveness of an assessment to low vision rehabilitation. Haymes and colleagues (2001) determined in a study with 22 participants that the Melbourne Low Vision ADL Index was responsive

Figure 2. Person–item map for the visual field or scotoma construct between the pretest (left) and posttest (right).

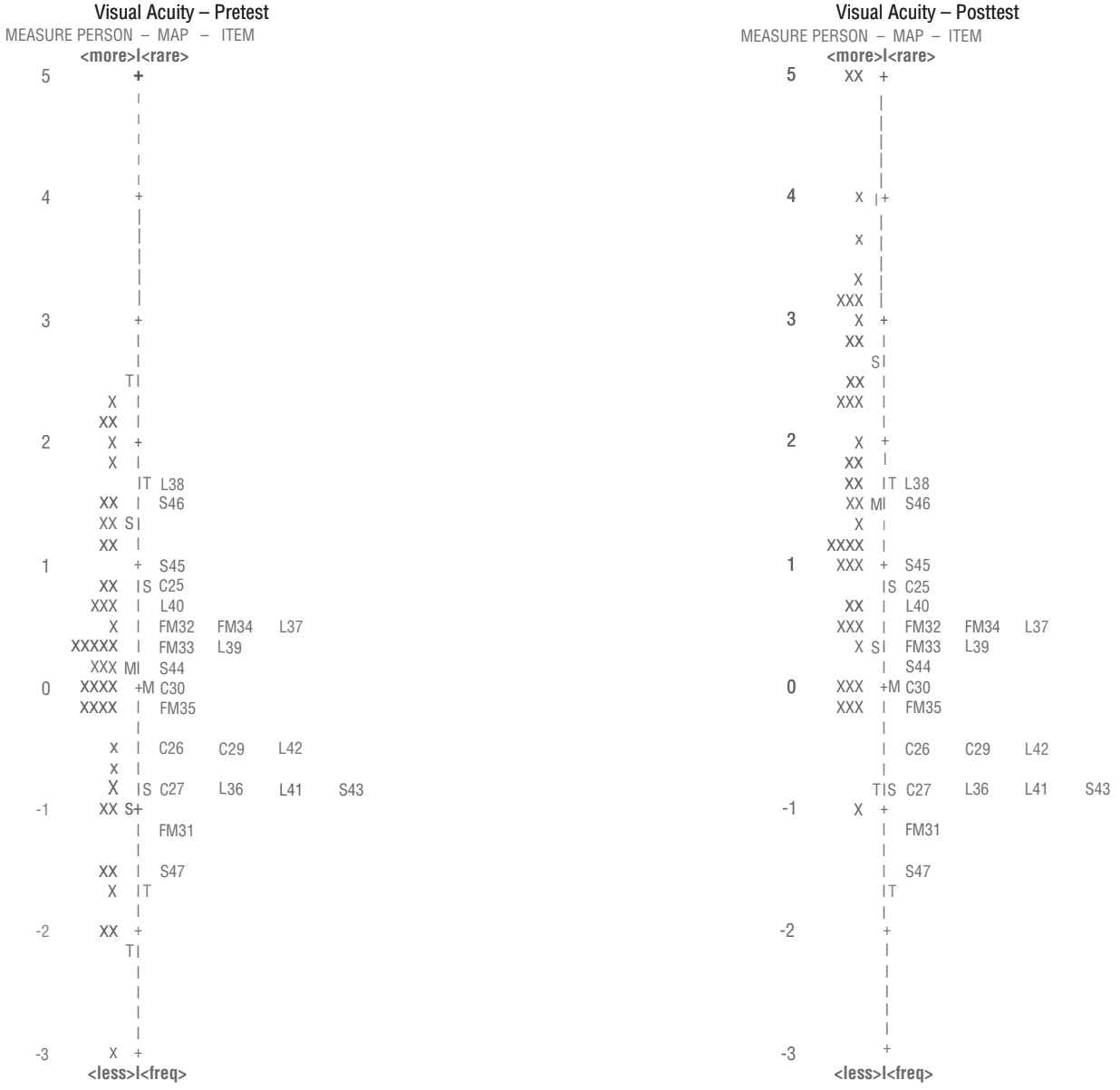


Note. Numbers on the left side indicate the measures of person ability and item difficulty in logits. Each X on the left side of the map indicates 1 participant; items are located on the right side of the map. The M to the left of the vertical line is the mean of the person abilities, and the M to the right of the vertical line is the mean of the item difficulties. Ss indicate 1 standard deviation and Ts indicate 2 standard deviations for item difficulty and person ability, respectively. C = Communication, FM = Financial Management, FP = Food Preparation, HM = Home Management, L = Leisure, M = Mobility; S = Shopping, SC = Self-Care.

(ES = 0.78) to changes after a low vision rehabilitation program for clients with macular degeneration. Administration of the Melbourne Low Vision ADL Index requires observation of the performance of 16 complex IADLs and ratings on nine questionnaire ADL items, making it time intensive to administer.

Stelmack et al. (2006) evaluated the sensitivity to change of the Veterans Affairs (VA) Low Vision Visual Functioning Questionnaire (LV VFQ–48) for 285 veterans who underwent vision rehabilitation. They found ESs of 1.90 for veterans who received inpatient rehabilitation and 0.29 for those who received outpatient rehabilitation.

Figure 3. Person–item map for the visual acuity construct between the pretest (left) and posttest (right).



Note. Numbers on the left side indicate the measures of person ability and item difficulty in logits. Each X on the left side of the map indicates 1 participant; items are located on the right side of the map. The M to the left of the vertical line is the mean of the person abilities, and the M to the right of the vertical line is the mean of the item difficulties. Ss indicate 1 standard deviation and Ts indicate 2 standard deviations for item difficulty and person ability, respectively. C = Communication, FM = Financial Management, FP = Food Preparation, HM = Home Management, L = Leisure, M = Mobility; S = Shopping, SC = Self-Care.

Stelmack and colleagues (2008, 2017) have documented improvements in participant visual ability in different VA settings, in response to different levels of intervention, and for participants with varying levels of visual acuity (Rubin, 2017). Approximately 90% of participants in Stelmack and colleagues' studies were White men with macular degeneration. Clients seen for low vision rehabilitation at the VA are provided prescribed optical devices at no cost and with no

expense spared in types of equipment dispensed (Rubin, 2017). Responsiveness to change of the LV VFQ–48 has been established only for VA participants.

Massof et al. (2007) developed the Activity Inventory, a visual function questionnaire, to be administered adaptively through a computer-assisted interview. The Activity Inventory is well validated (Goldstein et al., 2014) and is used frequently to measure the effectiveness of low vision rehabilitation. However, it contains 510 items used in querying difficulty in performance and importance to the client, and a computer is needed to administer it. In addition, this assessment is time intensive to administer.

Study Limitations

This study has several limitations. First, we did not gather data on the interventions used by the collaborating occupational therapists, who had different levels of experience, or on variations in interventions across settings. Second, all participants were rated at baseline as having good rehabilitation potential; results may differ for clients with other levels of rehabilitation potential. In addition, we collected no information on participants' visual skills at the client factor or performance skill level. Third, we used bootstrapping to account for the small sample size, and we were unable to determine minimal clinically important difference (MCID). Moreover, although we found some differences in person measures across settings, we could not evaluate differential item functioning across settings because of the small sample size and the risk of inflating Type II error rates. Finally, the implications of our findings are limited to the study participants and are not generalizable to other people with different levels of visual ability.

Implications for Occupational Therapy Practice

The findings of this study have the following implications for occupational therapy practice:

- The LVIM–R is responsive to change in visual ability resulting from occupational therapy for clients with a range of visual diagnoses resulting in low vision.
- The LVIM–R is useful for measuring occupational therapy outcomes in low vision rehabilitation.
- The LVIM–R is responsive to change for occupational therapy provided in both home health and outpatient settings.

Conclusion

The LVIM–R is responsive to changes in clients' visual ability after low vision rehabilitation provided by occupational therapists and can be used in settings including home health and outpatient clinics. The instrument's two distinct vision constructs cover a wide spectrum of visual ability related to ADLs and IADLs. Future studies of the LVIM–R are needed to determine the MCID, to compare the results of intervention and control groups, and to demonstrate which interventions are most effective in various settings and with clients of varying levels of visual ability.

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